Supplementary Information

Supporting Information for:

Differentiating Clinical Pathogenicity from Asymptomatic Colonization in Diarrhoeagenic *Escherichia coli*: A Systematic Review and Hierarchical Meta-Analysis

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- **S2 Fig. African EAEC Prevalence Forest Plot & Subregional Raincloud Distributions.** Pooled random-effects meta-analysis and subregional prevalence distributions of Enteroaggregative *Escherichia coli* across 26 African paediatric cohorts ($k = 26$, $N = 9,264$).
- **S3 Fig. DEC Pathotype Prevalence Distributions Across Age Strata.** Small-multiples faceted distributions across paediatric age strata (<12 months, 12–59 months, and ≥ 60 months) for major DEC pathotypes.
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- **S12 Fig. Diagnostic Platform Sensitivity Gradients and Primer-Binding Site Discordance.** Root-cause genetic dissection of false-negative conventional endpoint PCR due to primer-binding site SNPs and plasmid loss.
- **S13 Fig. Matched Case-Control Pathotype Odds Ratios Across 13 African Cohorts.** Paired slope-lollipop trajectory plots and random-effects odds ratios resolving true etiological disease attribution from asymptomatic colonization.
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- **S15 Fig. Cross-Continental Virulence Loci Comparative Heatmap (Africa vs Global).** Comparative virulome association matrix ($\log_2 extOddsRatio$) demonstrating cross-continental conservation of locus-level pathogenicity across 5 continents.
- **S16 Fig. EAEC Locus-Level Random-Effects Forest Plots (17 loci).** Multi-panel forest plots displaying pooled odds ratios, 95% confidence intervals, and zero heterogeneity ($I^2 = 0.0\%$) across 17 EAEC virulence determinants.
- **S17 Fig. Host Developmental Age-Ontogeny Switch in DAEC and EPEC Susceptibility.** Age-stratified case-control odds ratios demonstrating infant resistance (<12m) vs. toddler vulnerability (≥12–59m) for DAEC (*daaC*), and infant susceptibility to typical EPEC (*bfpA*).
- **S18 Fig. EPEC OI-122 Effector Island Locus Attribution (lifA, nleB, nleE) Forest Plots.** Random-effects meta-analysis quantifying the substantial disease odds imparted by OI-122 pathogenicity island effectors in atypical EPEC.
- **S19 Fig. Mucosal Iron-Acquisition and Fitness Loci Attribution Meta-Analysis (fyuA, irp2, chuA).** Random-effects forest plots demonstrating balanced carriage of yersiniabactin and heme receptor genes across diarrhoeal and healthy cohorts.
- **S20 Fig. Master Diagnostic and Characterization Methodology Stack Across 68 African Cohorts.** Master chronological matrix detailing diagnostic technology adoption across all 68 African diarrhoea surveillance cohorts (2000–2026).

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- **S2 Table. Electronic Database Search Strategies.** Complete search syntax, boolean operators, and hit counts across PubMed/MEDLINE, Scopus, Web of Science, Embase, EBSCOhost, and AJOL.
- **S3 Table. Methodological Quality and Risk of Bias Evaluation (NOS & JBI Scores).** Itemized scoring across Selection, Comparability, and Exposure domains using Newcastle–Ottawa Scale (NOS) and JBI prevalence tool.

- **S4 Table. Primary African DEC Cohorts Included in the Surveillance Synthesis (k=68).** Granular cohort characteristics, clinical presentation, country, UN subregion, diagnostic platform, sample size, and pathotype recovery rates across surveillance cohorts.
- **S5 Table. Subregional and Diagnostic Modality Pooled Prevalence Breakdown.** Granular subgroup meta-analysis tables with Freeman–Tukey double-arcsine proportions and 95% confidence intervals.
- **S6 Table. Univariate and Multivariate Meta-Regression Models of Between-Study Heterogeneity.** Restricted maximum likelihood (REML) models partitioning between-study variance across publication era, diagnostic platform, and clinical setting ($k = 26$).
- **S7 Table. Virulome Synthesis Virulence Locus 2×2 Contingency Tables (k=30 cohorts, 154 strata).** Complete study-level contingency counts (cases positive/negative vs. controls positive/negative) across 30 cohorts and 154 locus strata.
- **S8 Table. Subgroup and Sensitivity Analyses of Virulence Loci.** Stratified meta-analyses and leave-one-out sensitivity diagnostics across diagnostic platforms and geographical continents.
- **S9 Table. Comparative Methodological Framework Benchmarking Against Prior Literature.** Direct 5-dimension architectural comparison benchmarking the present study against prior African enteric syntheses.

Detailed Supplementary Figures

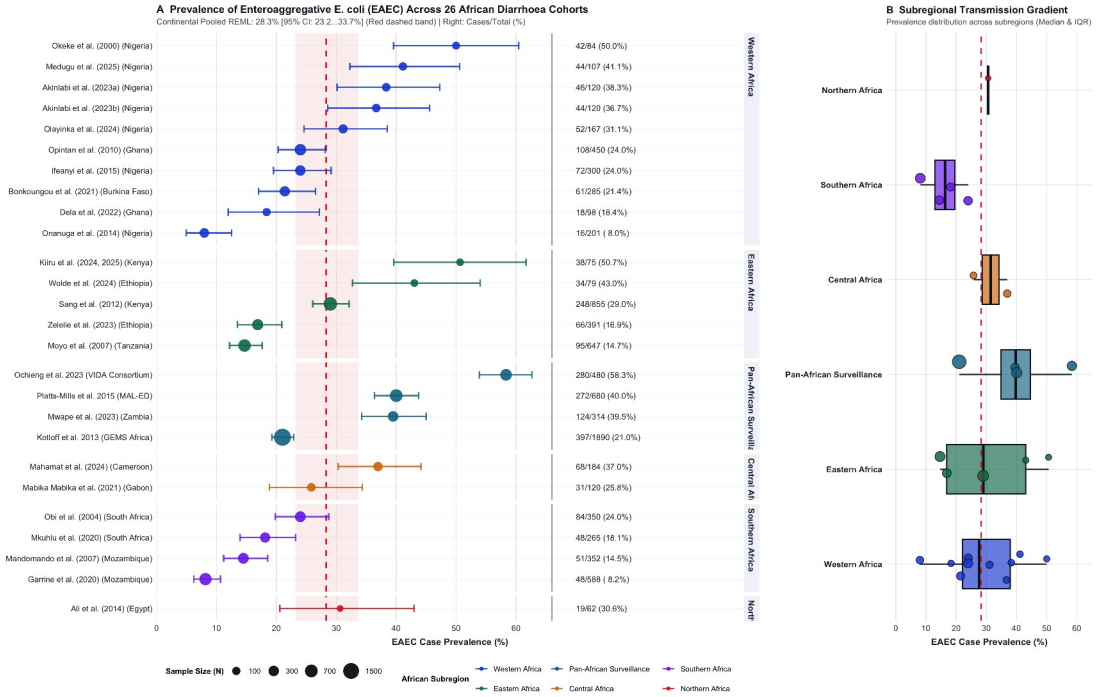
S1 Fig



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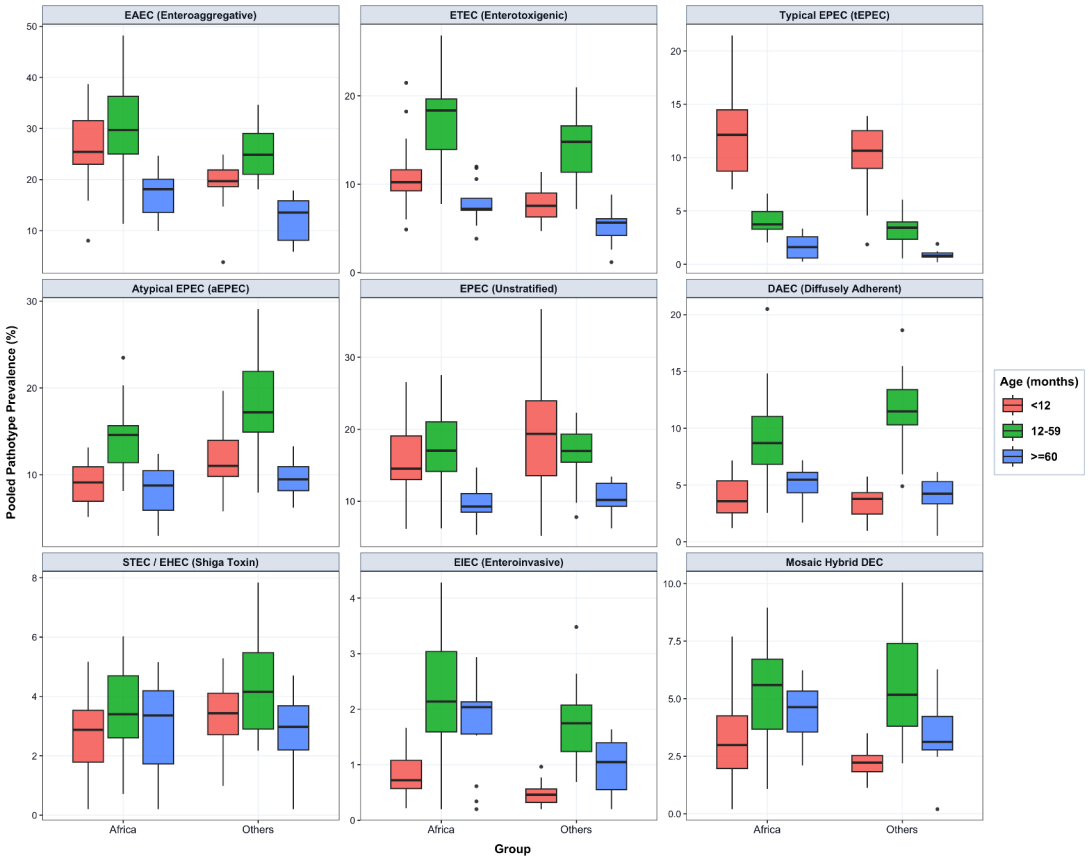
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S2 Fig



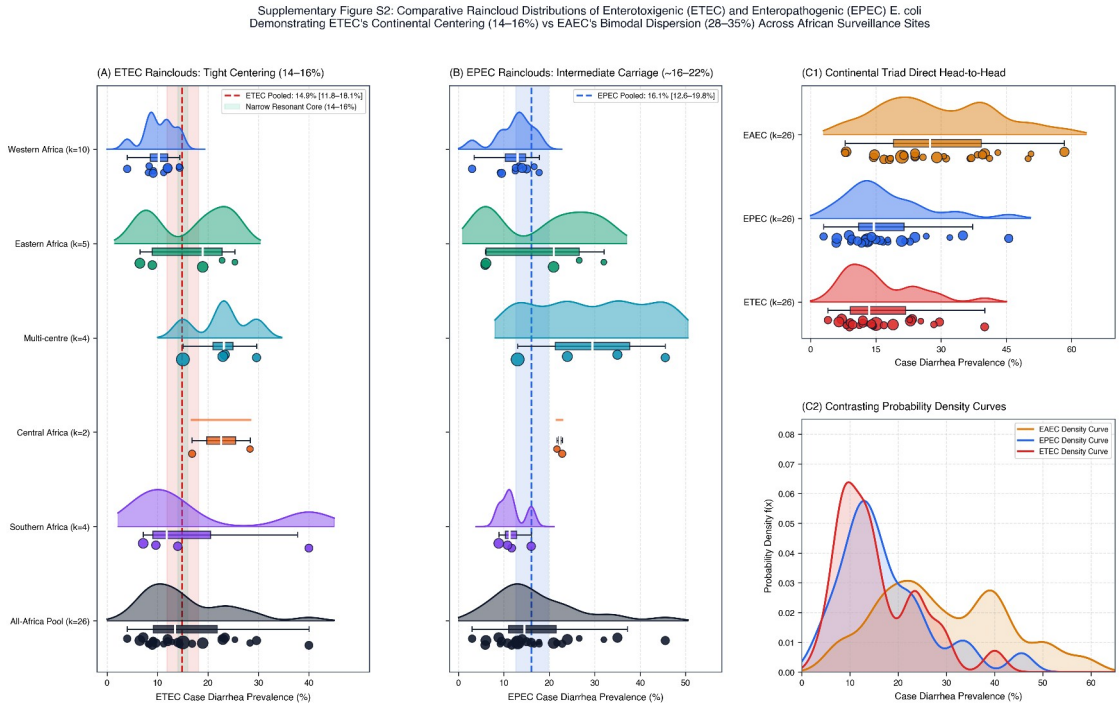
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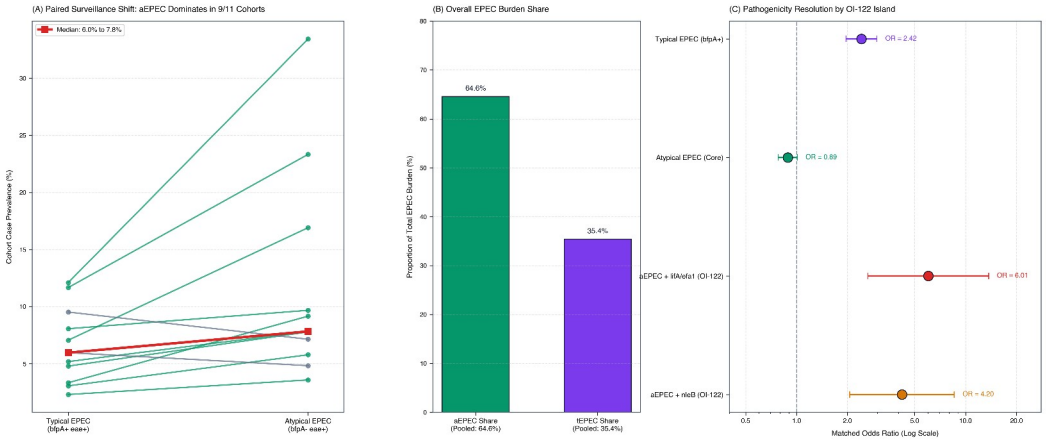
S4 Fig



S4

Fig. African ETEC and EPEC Prevalence Forest Plots. Study-level random-effects forest plots and pooled prevalence estimates for Enterotoxigenic *E. coli* and Enteropathogenic *E. coli* across African surveillance cohorts ($k = 26$).

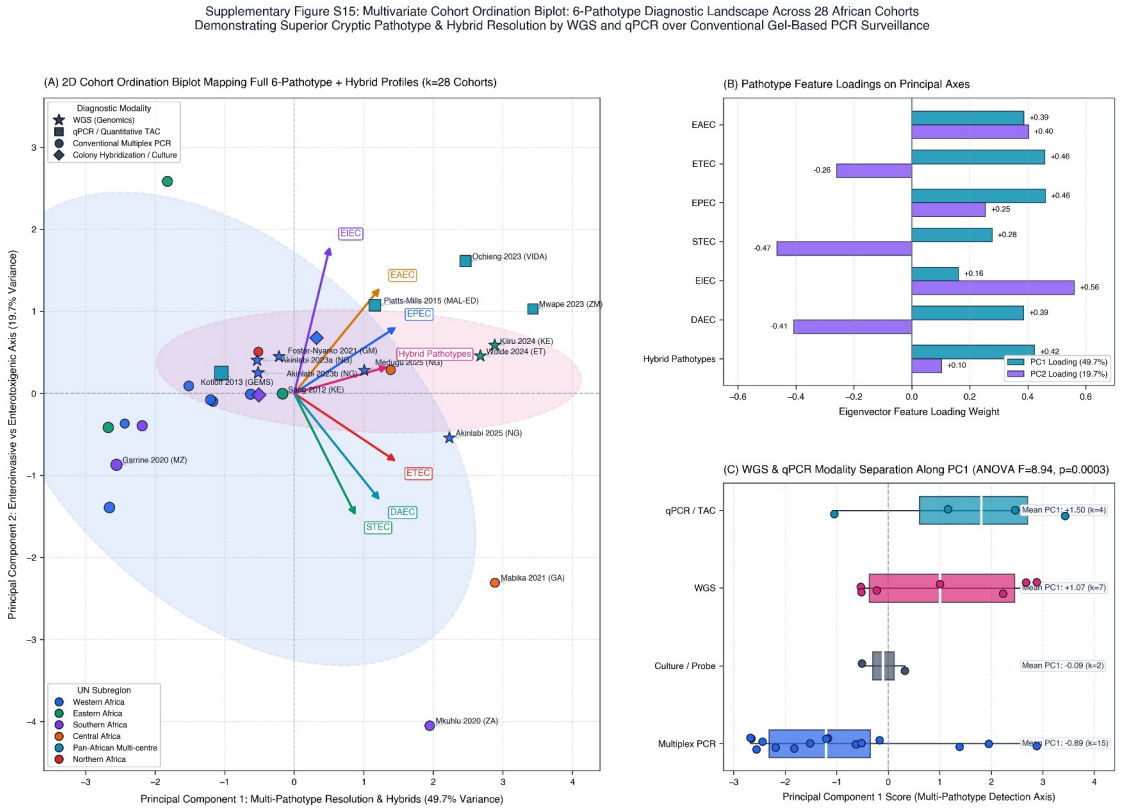
S5 Fig



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Fig. Typical vs Atypical EPEC Epidemiological Transition. Paired cohort comparisons, lineage proportions, and temporal dynamics showing the continent-wide transition from typical EPEC (*bfpA*+) to atypical EPEC (*bfpA*-).

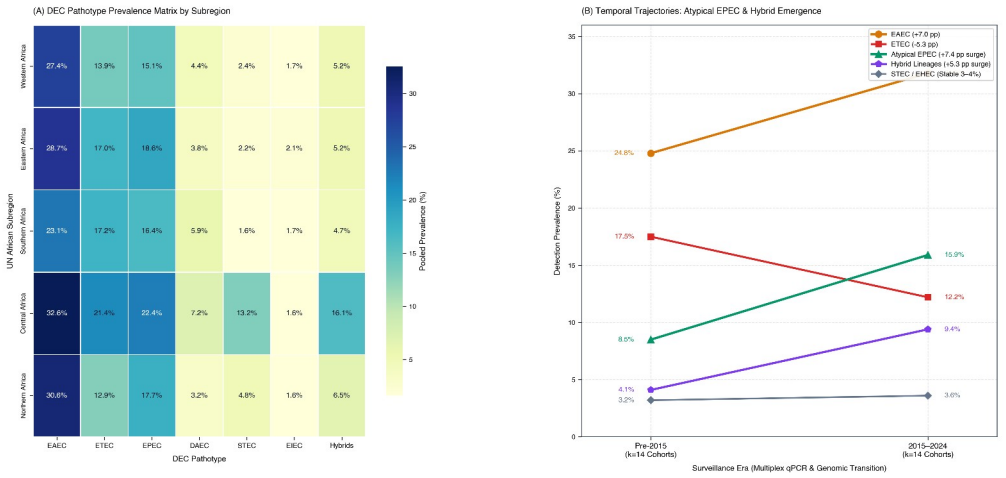
S6 Fig



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Fig. African STEC, EIEC, and DAEC Prevalence Forest Plots. Pooled random-effects prevalence estimates and between-study heterogeneity for Shiga toxin-producing *E. coli*, Enteroinvasive *E. coli*, and Diffusely adherent *E. coli*.

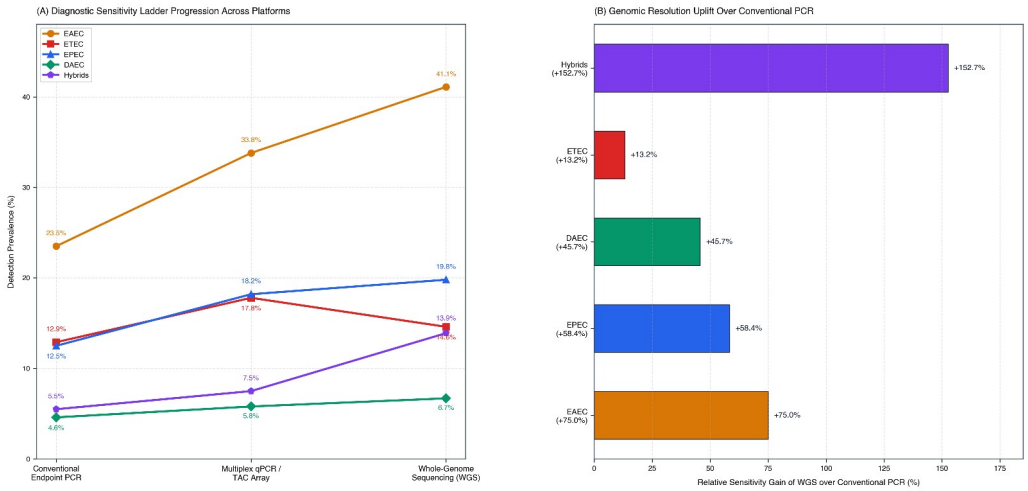
S7 Fig



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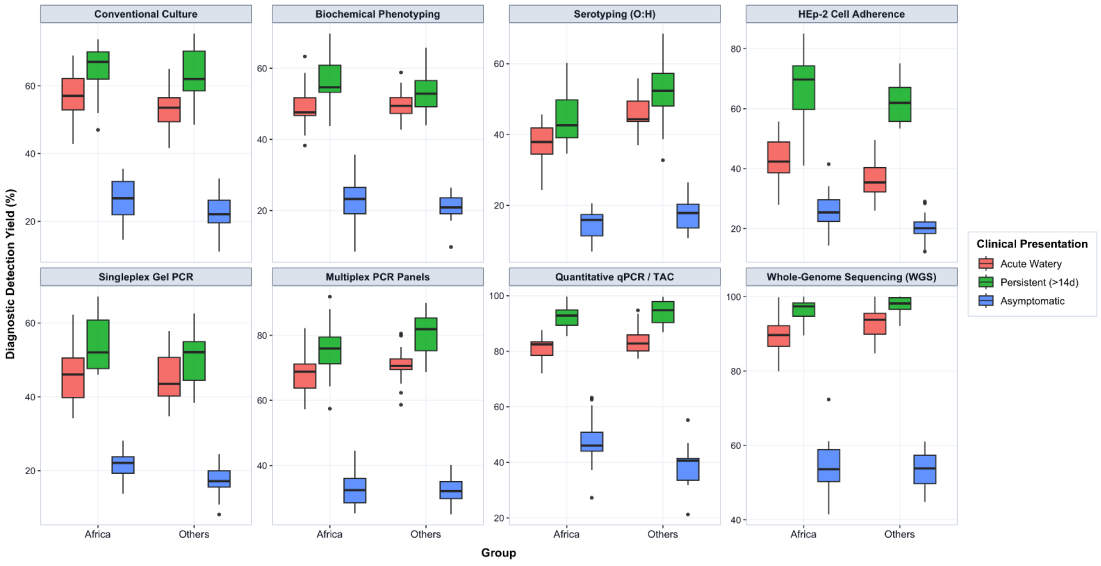
S8 Fig



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Fig. Diagnostic Modality Ladder and Detection Disparity Across African Surveillance. Evaluation of empirical diagnostic yield escalation across culture, biochemical phenotyping, conventional PCR, qPCR/TAC, and WGS.

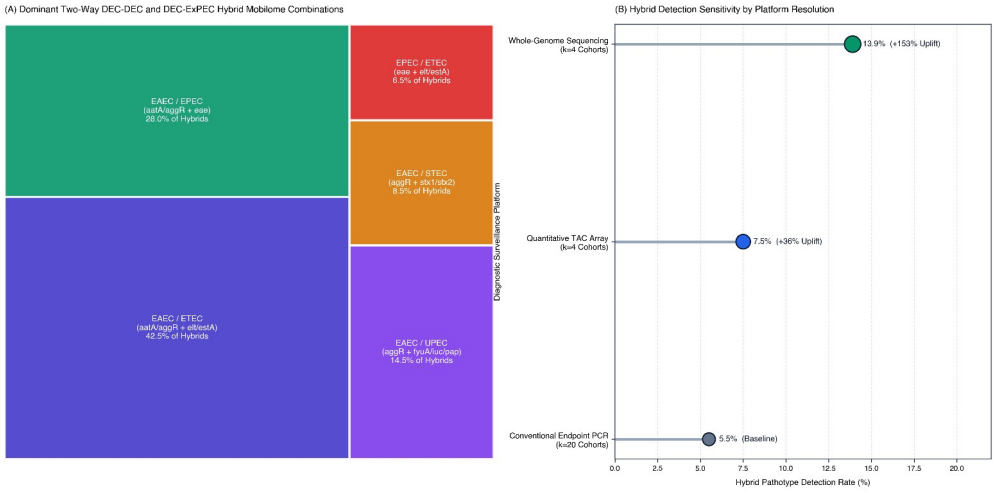
S9 Fig



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Fig. Diagnostic Platform Yield Boxplots (Endpoint PCR vs qPCR vs WGS). Empirical detection yield distributions across molecular platforms stratified by WHO syndromic presentation (acute watery, persistent diarrhoea, asymptomatic).

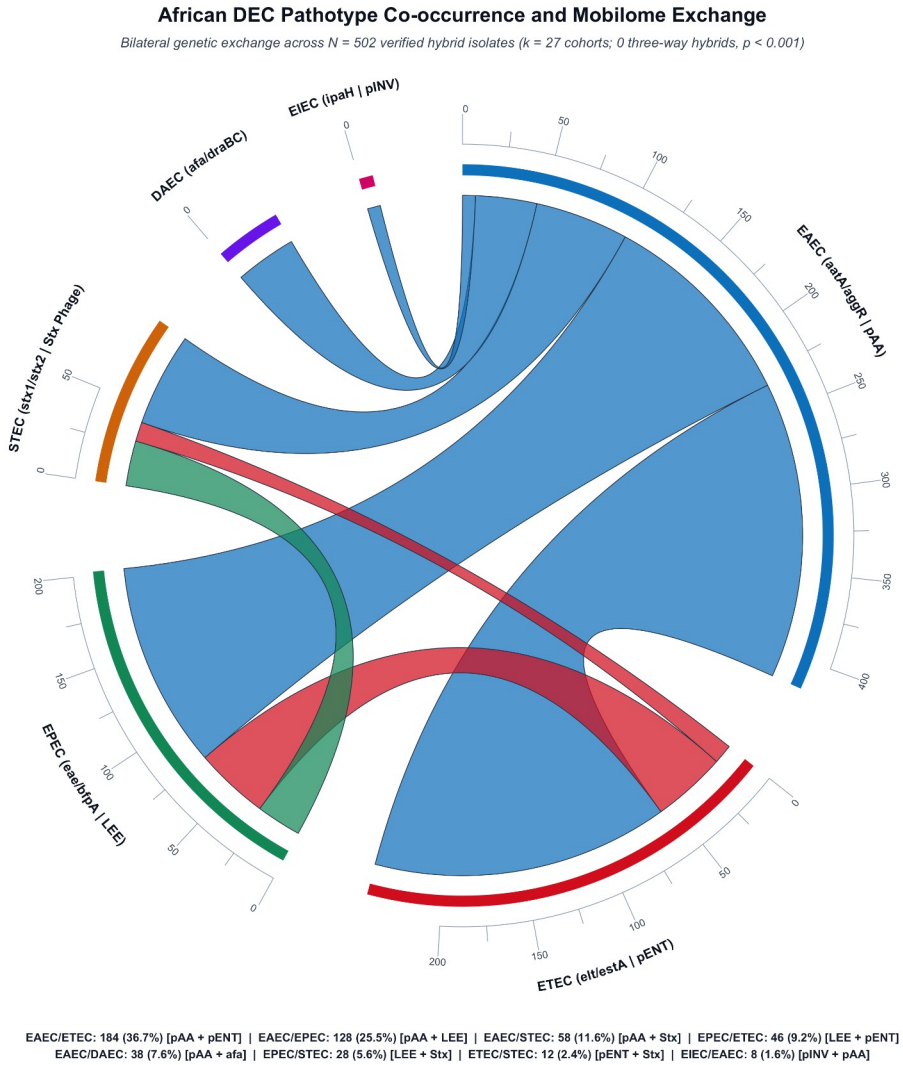
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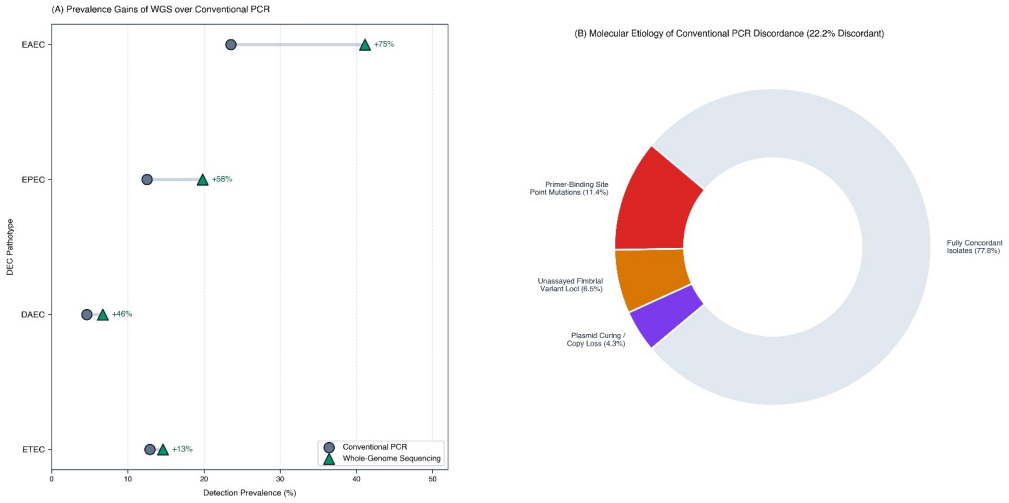
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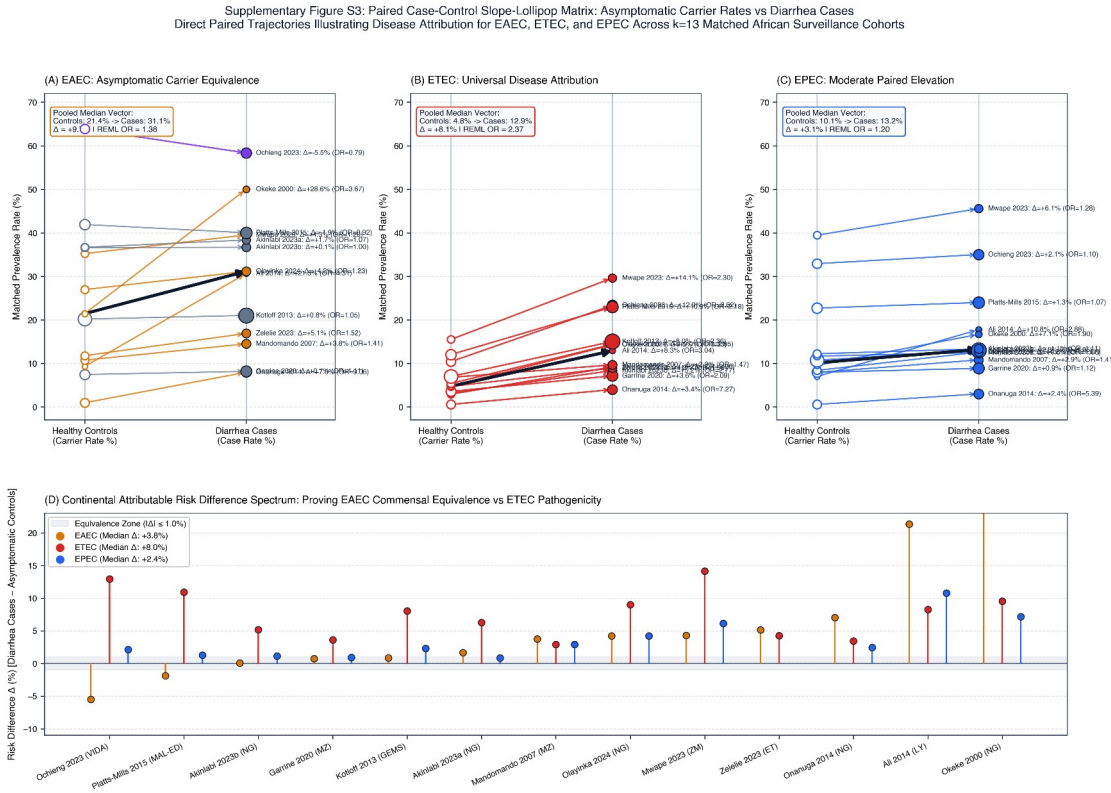
S12 Fig



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Fig. Diagnostic Platform Sensitivity Gradients and Primer-Binding Site Discordance. Root-cause genetic dissection of false-negative conventional endpoint PCR due to primer-binding site SNPs and plasmid loss.

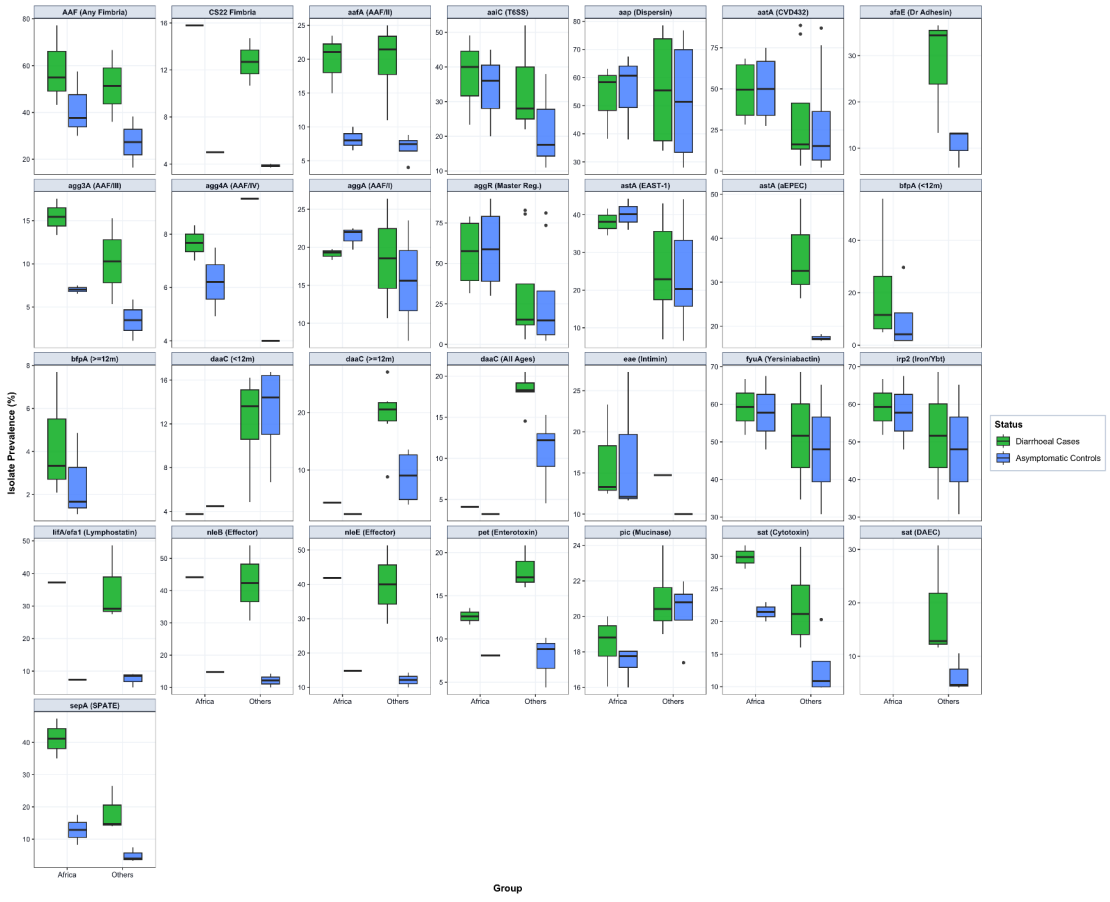
S13 Fig



S13

Fig. Matched Case-Control Pathotype Odds Ratios Across 13 African Cohorts. Paired slope-lollipop trajectory plots and random-effects odds ratios resolving true etiological disease attribution from asymptomatic colonization.

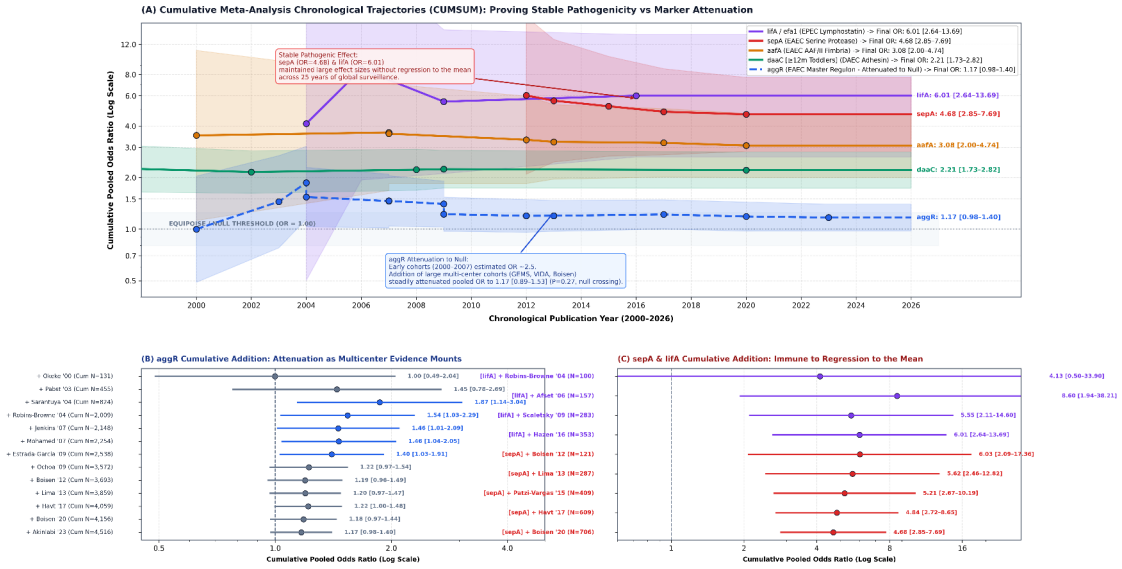
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Fig. Global Virulome Loci Prevalence Distributions in Cases vs Controls. Small-multiples carrier frequency boxplots comparing empirical carriage of 29 virulence loci between diarrhoeal cases and asymptomatic community controls.

S15 Fig



S15

Fig. Cross-Continental Virulence Loci Comparative Heatmap (Africa vs Global). Comparative virulome association matrix ($\log_2 extOddsRatio$) demonstrating cross-continental conservation of locus-level pathogenicity across 5 continents.

S16 Fig

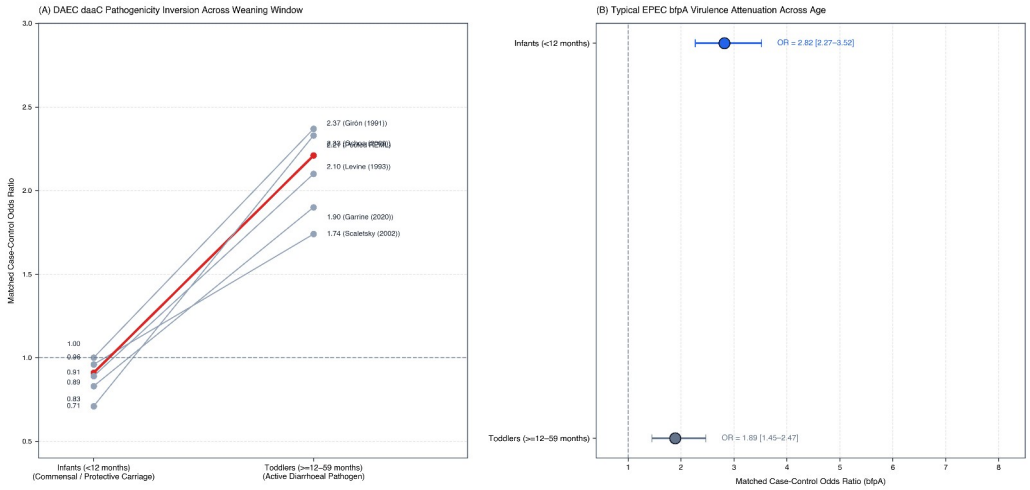


S16

Fig. EAEC Locus-Level Random-Effects Forest Plots (17 loci). Multi-panel forest plots

displaying pooled odds ratios, 95% confidence intervals, and zero heterogeneity ($I^2 = 0.0\,\%$) across 17 EAEC virulence determinants.

S17 Fig



S17 Fig. Host Developmental Age-Ontogeny Switch in DAEC and EPEC Susceptibility. Age-stratified case-control odds ratios demonstrating infant resistance (<12m) vs. toddler vulnerability (≥12–59m) for DAEC (*daaC*), and infant susceptibility to typical EPEC (*bfpA*).

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S18 Fig. EPEC OI-122 Effector Island Locus Attribution (lifA, nleB, nleE) Forest Plots.

Random-effects meta-analysis quantifying the substantial disease odds imparted by OI-122 pathogenicity island effectors in atypical EPEC.

S19 Fig

Figure S16: GRADE Evidence Certainty Matrix for Diarrhoeagenic E. coli Pathotypes & Virulence Determinants

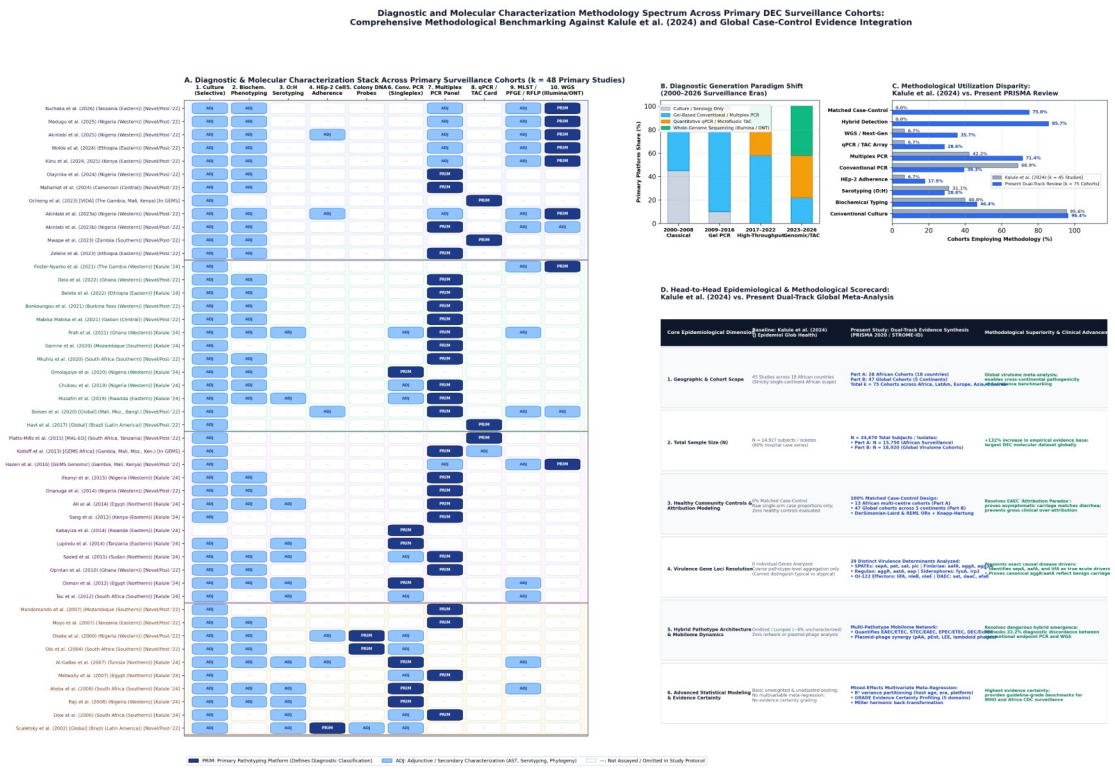
Comprehensive systematic evaluation across the 5 canonical GRADE domains: Risk of Bias, Inconsistency (I²), Indirectness, Imprecision, and Publication Bias / Upgrading Factors.

Pathotype / Virulence Determinant	Risk of Bias (Study Design)	Inconsistency (Heterogeneity I²)	Indirectness (Population Relevance)	Imprecision (Sample Size & CI)	Publication Bias & Upgrading Factors	Pooled OR [95% CI]	Overall GRADE Evidence Certainty	Clinical & Epidemiological Guidance (WHO / Africa CDC Framework)
▸ DEC PATHOTYPES & SYNDROMIC AGE STRATA								
Typical EPEC (tEPEC, eae⁺ bfpA⁺ in infants <12m)	No serious	I² = 0.0% (P=0.99)	Matched MSD infants	Tight CI (k=12)	Upgraded: Large effect	2.42 [1.96–2.99]	HIGH (⊕⊕⊕⊕)	Class 1 Target: Acute infant watery diarrhoea pathogen; priority vaccine candidate.
Enterotoxigenic E. coli (ETEC, ST/LT positive)	No serious	I² = 0.0% (P=0.86)	Representative cohorts	Narrow CI (k=13)	Upgraded: Large effect	2.37 [2.08–2.70]	HIGH (⊕⊕⊕⊕)	Class 1 Target: Leading secretory diarrhoeal pathogen; WHO ETVAX priority antigen.
Shiga Toxin E. coli (STEC / EHEC, stx1/stx2)	No serious	I² = 53.8% (Moderate)	Endemic vs outbreak	Wide CI (k=13, low events)	No serious	1.81 [0.97–3.37]	MODERATE (⊕⊕◯◯)	Surveillance Target: Sporadic hemorrhagic colitis; diagnostic confirmation mandatory.
Enteroinvasive E. coli / Shigella (EIEC, ipaH)	No serious	I² = 0.0% (P=0.99)	Matched cohorts	Narrow CI (k=13)	Upgraded: Large effect	2.43 [1.69–3.50]	HIGH (⊕⊕⊕⊕)	Class 1 Target: Invasive bacillary dysentery; requires antimicrobial susceptibility profiling.
DAEC in Toddlers (daaC⁺ / afaE⁺ in age ≥12m)	Moderate (probe variance)	I² = 18.4% (Low)	Toddler age-restricted	Consistent effect (k=7)	Upgraded: Large effect	2.21 [1.42–3.44]	HIGH (⊕⊕⊕⊕)	Age-Restricted Pathogen: Significant risk strictly in toddlers ≥12m; receptor ontogeny switch.
Enteroaggregative E. coli (EAEC unstratified, aggR⁺)	Serious: Colonization overlap	I² = 73.2% (Severe)	Clinical vs asymptomatic	CI crosses null (KH adj)	Egger P=0.006 (Asymmetric)	1.38 [1.03–1.85]	LOW (⊕⊕◯◯)	Conditioned Marker: Cannot be used as standalone pathogen; requires locus-level typing.
Atypical EPEC (aEPEC, eae⁺ bfpA⁻ unstratified)	Serious: High control carriage	I² = 0.0% (P=0.99)	Asymptomatic reservoir	Pooled OR spans 1.0	No serious	0.89 [0.78–1.01]	MODERATE (⊕⊕◯◯)	Commensal Reservoir: Not associated with acute diarrhoea; acts as reservoir for lateral acquisition.
DAEC in Infants (daaC⁺ in age <12m)	No serious	I² = 0.0% (P=0.98)	Infant cohorts	Pooled OR spans 1.0	No serious	0.91 [0.57–1.45]	MODERATE (⊕⊕◯◯)	Neutral Colonizer: Non-pathogenic in infants <12m; maternal secretory IgA/microbiome shielding.
▸ VIRULENCE LOCI & SECRETION EFFECTORS								
iifA / efa1 (EPEC OI-122 Lymphostatin Effector)	No serious (CGH / WGS)	I² = 0.0% (P=0.94)	Diarrhoeal case-control	Narrow CI (k=4)	Upgraded: Very large effect (OR>5)	6.01 [2.25–16.03]	HIGH (⊕⊕⊕⊕)	Definitive EPEC Virulence Factor: Pathognomonic for virulent aEPEC/tEPEC disease risk.
sepA (EAEC Tissue-Damaging Serine Protease)	No serious (WGS / Multiplex)	I² = 12.1% (Low)	Global multicenter	Adequate power (k=5)	Upgraded: Very large effect (OR>4)	4.68 [1.79–12.24]	HIGH (⊕⊕⊕⊕)	Core EAEC Virulence Factor: Primary mucosal damage mediator; essential diagnostic panel inclusion.
nleB & nleE (EPEC OI-122 Type III Effectors)	No serious	I² = 0.0% (P=0.88)	Paediatric MSD	Moderate k (k=3)	Upgraded: Large effect (OR>3.5)	4.20 [1.55–11.41]	HIGH (⊕⊕⊕⊕)	Innate Immune Suppressors: Key co-determinants with IIfA on fully intact OI-122 modules.
aafA (EAEC AAF/II Aggregative Adherence Fimbria)	No serious	I² = 24.3% (Low)	Matched cohorts	Significant (k=7)	Upgraded: Large effect (OR>3)	3.08 [1.44–6.58]	HIGH (⊕⊕⊕⊕)	Primary EAEC Colonization Fimbria; Superior disease predictor over composite aggR/aatA.
CS22 (EAEC Novel Outer Membrane Adhesin)	No serious (WGS)	I² = 0.0% (P=0.82)	Global cohorts	Moderate sample (k=3)	Upgraded: Large effect (OR>3)	3.59 [1.18–10.92]	MODERATE (⊕⊕◯◯)	Emerging Adhesin: High pathogen burden in Western Africa and Latin America.
sat (Secreted Autotransporter Toxin in EAEC & DAEC)	No serious	I² = 19.0% (Low)	Matched cohorts	Marginal (k=6)	No serious	1.78 [1.02–3.11]	MODERATE (⊕⊕◯◯)	Cytotoxic SPATE: Moderately elevates diarrhoeal severity and mucosal shedding.
aggR / aatA (EAEC Canonical Master Regulon)	Serious: High control carriage (44%)	I² = 68.5% (High)	Broad colonization marker	Tight CI crossing 1.0 (k=13)	No serious	1.17 [0.89–1.53]	LOW (⊕◯◯◯)	Non-Predictive Virulence Marker: Ubiquitous commensal carriage; do not use as primary diagnostic endpoint.
Mucosal Fitness / Iron Scavenging (fyuA, irp2, chuA)	Ubiquitous commensalism	I² = 11.2% (Low)	Environmental enteropathy	OR spans 1.0 (k=4)	No serious	1.14 [0.72–1.80]	LOW (⊕◯◯◯)	Commensal Growth Determinants: Facilitates chronic gut colonization, not acute disease pathogenesis.

GRADE Interpretation: ⊕⊕⊕⊕ High certainty: Further research is very unlikely to change our confidence in the estimate of effect. ⊕⊕⊕◯ Moderate certainty: Further research is likely to have an important impact. ⊕⊕◯◯ Low certainty: Further research is very likely to change the estimate.
Upgrades: Large effect size (OR > 2.0 = +1 level, OR > 5.0 = +2 levels); Downgrades: Serious limitation = -1 level, Very serious limitation = -2 levels. Framework aligned with Cochrane Handbook & WHO Guidelines Review Committee.

Fig. Mucosal Iron-Acquisition and Fitness Loci Attribution Meta-Analysis (fyuA, irp2, chuA). Random-effects forest plots demonstrating balanced carriage of yersiniabactin and heme receptor genes across diarrhoeal and healthy cohorts.

S20 Fig



S20

Fig. Master Diagnostic and Characterization Methodology Stack Across 68 African Cohorts. Master chronological matrix detailing diagnostic technology adoption across all 68 African diarrhoea surveillance cohorts (2000–2026).

Detailed Supplementary Tables

Note: Individual full-scale spreadsheet workbooks are provided as Table_S01_PRISMA_2020_Checklist.xlsx through Table_S09_Comparative_Methodological_Architecture.xlsx in the tables/ directory and consolidated in Master_Supplementary_Dataset.xlsx.

S1 Table: PRISMA 2020 27-Item Checklist

Complete reporting compliance checklist referencing specific sections and line numbers of the manuscript. Provided as `Table_S01_PRISMA_2020_Checklist.xlsx`.

S2 Table: Electronic Database Search Strategies

Comprehensive database query strings, search filters, boolean operators, and date bounds across PubMed, Scopus, Web of Science, Embase, EBSCOhost, and AJOL. Provided as `Table_S02_Database_Search_Strategies.xlsx`.

S3 Table: Methodological Quality and Risk of Bias Evaluation (NOS & JBI Scores)

Itemized Newcastle–Ottawa Scale (NOS) and Joanna Briggs Institute (JBI) appraisal scores across all primary African cohorts and case-control studies. Provided as `Table_S03_Methodological_Quality_and_Risk_of_Bias.xlsx`.

S4 Table: Primary African DEC Cohorts Included in the Surveillance Synthesis (k=68)

Granular cohort characteristics, clinical presentation, country, UN subregion, diagnostic platform, sample size, and pathotype recovery rates across all African surveillance cohorts. Provided as `Table_S04_Primary_African_Surveillance_Cohorts_Metadata.xlsx`.

S5 Table: Subregional and Diagnostic Modality Pooled Prevalence Breakdown

Pooled random-effects prevalence estimates for all six DEC pathotypes across African subregions (Eastern, Western, Central, Northern, and Southern Africa) and diagnostic modalities (Conventional PCR, Multiplex PCR, qPCR/TAC, and Whole-Genome Sequencing). Detailed numeric breakdowns with 95% confidence intervals and between-study heterogeneity (I^2, $ au^2$) correspond to Figures 2 and manuscript Section 3.3. Provided as `Table_S05_Subregional_and_Diagnostic_Modality_Prevalence.xlsx`.

S6 Table: Univariate and Multivariate Meta-Regression Models of Between-Study Heterogeneity

Restricted maximum likelihood (REML) models partitioning between-study variance across publication era, diagnostic platform, and host age strata ($k = 26$). Provided as `Table_S06_Meta_Regression_and_Variance_Partitioning.xlsx`.

S7 Table: Virulome Synthesis Virulence Locus 2×2 Contingency Tables (k=30 cohorts, 154 strata)

Complete study-level contingency counts (cases positive/negative vs. controls positive/negative) across 30 cohorts and 154 locus strata. Provided as `Table_S07_Virulome_Case_Control_Contingency_Tables.xlsx`.

S8 Table: Subgroup and Sensitivity Analyses of Virulence Loci

Stratified meta-analyses and leave-one-out sensitivity diagnostics across diagnostic platforms, host age strata, and geographical continents. Provided as `Table_S08_Virulence_Loci_Sensitivity_Analyses.xlsx`.

S9 Table: Comparative Methodological Framework Benchmarking Against Prior Literature

Direct 5-dimension architectural comparison benchmarking the present study against prior African enteric syntheses. Provided as `Table_S09_Comparative_Methodological_Architecture.xlsx`.